CS Chapter 5 Review - Critical Thinking Questions

1. What is the purpose of a loop structure?
A loop structure allows a program to repeat a block of code automatically multiple times instead of manually writing out duplicate statements.

2. Explain the difference between a while statement and a do-while statement.
A while loop checks its condition before running the code block, so if the condition is false initially, the code inside never runs. A do-while loop executes its code block at least once before checking the condition at the end.

3. In which review of this chapter did you write code for an input validation loop?
We wrote input validation loops in our guessing game and menu-driven programs where the code keeps prompting the user until they enter a valid option or correct range.

4. a) What is an infinite loop?
An infinite loop runs endlessly because its exit condition never evaluates to false.

b) List two types of errors that can lead to an infinite loop.
- A logic error where the loop control variable is updated in the wrong direction (e.g., subtracting instead of adding).
- Omitting the step that updates the loop control variable inside the loop body (like forgetting i++).

c) What is meant by overflow?
Overflow occurs when a variable's value exceeds the maximum limit supported by its data type, causing it to exceed storage boundaries.

5. How many times will the do-while loop execute?
It will execute 60 times. Since x starts at 0 and increases by 2 each iteration, it takes 60 loops for x to reach 120 and stop the loop (120 / 2 = 60).

6. What initial value of x would make the loop infinite?
Any initial value less than 120 (for example, 0 or any negative number). Because the loop subtracts 3 each time, the value moves further away from 120, keeping x < 120 true forever.

7. Compare and contrast counters and accumulators. List two uses for each.

- Compare: Both are variables used in loops to keep a running tally across iterations.
- Contrast: A counter increases by a fixed step (typically +1) to track how many times an event occurs, whereas an accumulator increases by varying amounts to maintain a cumulative sum.
- Uses for Counters:
  1. Tracking the number of invalid login attempts.
  2. Counting how many items meet a specific condition in a list.
- Uses for Accumulators:
  1. Summing up item prices for a shopping cart total.
  2. Accumulating test scores to calculate a class average.

8. Write a for statement that sums the integers from 3 to 10, inclusive.

```
for (int i = 3; i <= 10; i++) {
    sum += i;
}
```

9. List two factors that should be considered when determining which loop structure to choose.
- Whether the loop needs to run at least once regardless of conditions (suggesting a do-while loop).
- Whether the exact number of iterations is known ahead of time (making a for loop ideal) or depends on an unknown user input/condition (making a while loop better).

11. Consider String x = "my string."; Determine the value returned by each method:
a) x.length() → 10 (Counts all letters, the space, and the period).
b) x.substring(0, 3) → "my " (Extracts characters from index 0 up to, but not including, index 3).